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Accelerating digital transformation: key drivers of cloud computing adoption and firm performance outcomes in Malaysian SMEs

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Abstract

This study investigates the determinants influencing cloud computing adoption among small- and medium-sized enterprises (SMEs) in the Klang Valley, Malaysia, a region representing 46% of national SME concentration and contributing significantly to Malaysia's digital economy hub. Despite the recognized benefits of cloud computing, Malaysian SMEs lag behind global and regional counterparts in adoption rates. This study advances existing literature by integrating four theoretical perspectives—Technology–Organization–Environment (TOE), Diffusion of Innovation (DOI), Practice-Based View (PBV), and Dynamic Capabilities (DC)—to develop a comprehensive multi-level framework that addresses the limitations of single-theory approaches. Using a purposive sampling strategy, which while limiting generalizability, enables targeted investigation of SMEs actively evaluating cloud technologies, this research surveyed 350 SMEs to identify key factors driving cloud adoption. Data cleaning procedures included outlier detection, missing value analysis, and normality testing using both univariate and multivariate approaches. Validation steps encompassed content validity through expert review, construct validity via confirmatory factor analysis, and reliability testing through Cronbach's alpha and composite reliability measures. The study employed structural equation modeling using SmartPLS to analyze relationships between variables. Results reveal that relative advantage, security assurance, industry influence, competitive pressure, corporate culture, and IT competency significantly influence cloud computing adoption. The non-significance of cost-effectiveness ($\beta = 0.054$, $p = 0.164$) contradicts conventional wisdom but aligns with Malaysian SMEs' prioritization of capability over cost, possibly due to government subsidies and vendor promotional pricing. Similarly, the non-significance of complexity ($\beta = 0.042$, $p = 0.276$) suggests that Malaysian SMEs may lack the technical expertise to adequately assess system complexity, indicating a critical capability gap. This research makes three key theoretical contributions: (1) validates an integrated multi-theory framework demonstrating complementarity between TOE-DOI for adoption factors and PBV-DC for performance outcomes; (2) establishes top management support as a critical mediating mechanism (explaining 41.9% of variance); and (3) identifies vendor support as a boundary condition strengthening the adoption-performance relationship. Most importantly, cloud computing adoption strongly influences firm performance, explaining 63.5% of performance variance. These findings provide specific, actionable insights including SME leaders should prioritize building internal IT competencies through structured training programs; vendors should offer tiered support packages tailored to SME maturity levels; and policymakers should implement sector-specific cloud adoption incentives beyond generic digitalization initiatives.

Keywords Cloud computing, Top management support, Vendor support, Firm performance, Malaysian SMEs, TOE framework, DOI theory

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Introduction

According to Research and Markets [1], the market size of worldwide cloud computing is projected to rise from USD 371.4 billion in 2020 to USD 832.1 billion by 2025, with an annual compound growth rate of 17.5% attributed to several factors, such as an increase in digitalization to deliver work-from-home readiness and cost-effectiveness purposes, especially during the pandemic period. According to another source, Allied Market Research [2], the cloud computing market was valued at USD 264.8 billion in 2019 and is anticipated to attain USD 927.5 billion by 2027, with a CAGR of 16.4%. This global trend presents both opportunities and challenges for emerging economies like Malaysia, where digital transformation remains uneven across enterprise segments.

According to the 2022 SME statistics, 97.4% of businesses in Malaysia are SMEs, comprising 1,173,601 establishments. Among the 97.4% of SMEs established, microenterprises have the largest share, at 78.7%, while small enterprises account for 19.7% and medium enterprises for 1.6%. In terms of Malaysian SMEs' performance in 2022, SMEs contributed 38.4% to the country's GDP, with 11.6% growth compared to the previous year [3]. The overall Malaysian GDP growth in 2022 was 8.7% [4], indicating that the GDP growth in the SME segment was higher than that of the non-SME segment, also known as the enterprise segment. The Klang Valley, encompassing Kuala Lumpur and surrounding Selangor state, hosts approximately 46% of these SMEs, making it Malaysia's economic epicenter and a critical context for understanding technology adoption patterns that may subsequently diffuse nationwide.

Total spending is often used as a straightforward metric to gauge the adoption rate of nascent technologies, including cloud computing. It offers a clearer view of acceptance levels, growth trends, and scale when coupled with the growth rate. According to a report by Twimbit [5], cloud computing spending in Malaysia accounts for approximately 3.63% of total IT spending, with a growth rate of 13.13% in public cloud computing. This positions Malaysia significantly below regional peers like Singapore (8.2%) and Thailand (5.1%), indicating substantial untapped potential. This positions Malaysia as a country with a relatively lower acceptance of public cloud computing [6]. However, there is a lack of focus on SMEs, especially in the Klang Valley, Malaysia. SMEs should seize the opportunity to leverage cloud computing technology to accelerate their businesses to another level due to the beneficial results that cloud computing brings to enhance firm performance, as the adoption of cloud computing has a significant influence on firm performance, which has been proven by prior research [7, 8]. The lag is particularly pronounced among SMEs, where resource constraints, capability gaps, and risk aversion create multiple adoption barriers.

From a theoretical perspective, existing research has predominantly applied single theoretical lenses to explain cloud adoption, resulting in fragmented understanding. TOE framework studies focus on contextual factors but neglect innovation characteristics. DOI research emphasizes innovation attributes but overlooks organizational capabilities. This study addresses these limitations by integrating multiple theoretical perspectives to provide a holistic understanding of the adoption–performance relationship. The study fills the gap of the TOE and DOI by including two firm-related theories and models, the theory of PBV and DC, to elucidate the connection between cloud computing adoption and firm performance, together with top management support and vendor support as mediating and moderating variables.

In terms of SMEs challenges from an industry point of view, three critical gaps emerge: (1) technology awareness gap—67% of Malaysian SMEs lack understanding of cloud computing benefits and implementation requirements; (2) capability gap—only 23% possess adequate IT competencies for cloud migration; and (3) trust gap—54% express security and vendor reliability concerns (SME Corp. Malaysia [9]). Prior research has underscored a knowledge deficit in Malaysia about the restricted implementation of cloud computing, attributed to inadequate investment in IT and a lack of understanding about the necessary actions for adoption [10]. Furthermore, the failure of Malaysian companies to adapt to the technology advancements of global corporations has exacerbated the competitiveness gap [10]. These gaps are exacerbated by the fragmented vendor landscape, where SMEs struggle to evaluate competing offerings and identify appropriate solutions.

This study makes four key contributions to address these gaps. First, it develops and validates an integrated theoretical framework combining TOE, DOI, PBV, and DC theories, demonstrating how different theoretical perspectives complement rather than compete in explaining adoption phenomena. Second, it provides empirical evidence from an understudied context—Malaysian SMEs in Klang Valley—contributing to the geographic diversification of cloud adoption literature. Third, it identifies the mediating role of top management support and moderating effect of vendor support, revealing critical boundary conditions for the adoption-performance relationship. Fourth, it offers actionable recommendations for practitioners and policymakers based on rigorous empirical analysis.

Conceptual background

Definition of key concepts

Innovative characteristics have five original characteristics, according to the DOI theory [11], which aims to elucidate how, why, and at what rate new ideas and technologies spread. Three out of five DOI constructions will be utilized in the study, which are complexity, relative advantage, and compatibility, and these findings are consistently supported and associated with the adoption of an innovation or technology [7, 12].

Technological context under TOE is technology, availability, and characteristics according to the TOE framework [13]. In the study, cost-effectiveness and security assurance are the sub-dimensions underneath the construct, besides innovative characteristics with relative advantage, compatibility, and complexity. Environment context under TOE encompasses industry features, market structure, technological support infrastructure, and governmental factors [13]. In the study, industry influence, competitive pressure, and regulatory compliance are the sub-dimensions underneath the construct, and environmental context is all about external factors. Organizational context under the TOE paradigm encompasses both formal and informal linking structures, communication procedures, organizational size, and resource slack [13]. In the study, corporate culture, and IT competency are the sub-dimensions underneath the construct, and organizational context is all about internal factors.

Top management support entails dedicating time to the IT program in proportion to its expenses and potential, encompassing the assessment of plans, oversight of outcomes, and the resolution of pertinent IT and business issues with management [14]. In Qasem et al.'s study [15], top management support pertains to the attitudes of senior management toward relevant technology, with the extent of support for the adoption and effective implementation of cloud computing dependent on the capabilities of top management to facilitate or influence it. In Li's study [16], management support denotes the management's readiness to embrace new technology through overall perception, investment rationale, proactive measures, and additional factors. Top management support refers to the extent to which an innovation receives endorsement and approval from the firm's senior management [17].

Vendor support is the level at which cloud computing vendors provide extended assistance and consultation to their clients [18]. Successful cloud-based technology applications require vendor support, given the essentiality of partnerships for inter-organizational system acceptance or adoption [19]. Vendor support is the level of trustworthiness and reliability of the solution or services provided by the cloud computing vendors [20], and a lack of such a factor may jeopardize or prolong the evaluation of the adoption of the technology. Vendor support is the degree to which the vendor provides support to the client while evaluating, testing, and selecting services [17]. Besides that, it also covers the vendor's ability to manage and deploy various tangible and intangible IT resources to provide the services [17].

Cloud computing adoption is defined as successfully implementing cloud computing technology within the organization at any scale by delivering certain benefits such as advantageous, better cost-effectiveness, easier to use, easier to scale, better reliability, more secure, better privacy, and so on. The level of cloud computing adoption can vary within an organization depending on the depth of the adoption. Organizations strategically employ cloud computing to save costs, hedge risks, and enhance the scalability of their database capabilities (HCL [21]). Cloud computing adoption is a strategic approach employed by organizations to enhance the scalability of information systems hosted on the Internet, while simultaneously minimizing expenses and risks [22], which means cloud computing is more advance, easy to use, more secure including better privacy [23, 24].

Firm performance in the study might be approached with objective metrics, including accounting data and financial indicators or ratios, as compared with financial performance, which is a business result measurement that may not depend solely on organization's promptitude but could be based on the industry which the organization run [25]. Khayer et al. [26] define firm performance as the superiority an organization obtains over its competitors by utilizing cloud computing in daily operations. Yoo and Kim [27] define firm performance as satisfaction, positive attitude, and consistency with expectations of cloud computing systems. The business environment has been evolving and growing increasingly competitive. Hence, firms have continued to improve their performance, including growth, profitability, return on investment, productivity, efficiency, effectiveness, and competitiveness [28].

Review of related theories and models

The TOE framework was delineated about the technological innovation processes [13], depicting the innovation process from conception to adoption and execution. The TOE framework is a theoretical paradigm that clarifies the influence of three distinct contexts—technological, organizational, and environmental—on the decision to adopt at the organizational level [29]. The elements within TOE's technological context include technology, availability, and characteristics. Meanwhile, industry characteristics and market structure, technology support

infrastructure, and government regulation constitute the environmental context.

DOI is extensively utilized in the study of technological adoption, which aims to elucidate the mechanisms, reasons, and pace at which new ideas and technologies are adopted. The development of this approach was undertaken by Everett Rogers and introduced in 1962. The primary attributes of innovation are relative advantage, compatibility, complexity, trialability, and observability [11]. Relative advantage refers to the degree to which a new technology is perceived as superior to its predecessor; compatibility denotes its congruence with contemporary values, historical experiences, and prerequisites; complexity denotes the degree of difficulty in understanding and utilizing it; trialability refers to the degree to which something can be tested on a restricted scale, whereas observability assesses the visibility of its results [11].

TOE and DOI were employed to elucidate the relationship between independent variables and the adoption of cloud computing. In contrast, these additional two theories will be utilized to elucidate the relationship between firm performance and the subsequent two concepts: Theory of Practice-Based View (PBV) and Dynamic Capabilities (DC). PBV elucidates the correlation among cloud computing adoption, vendor support, and firm performance within the study. The PBV model, developed by Bromiley and Rau [30], seeks to address the current research deficiency in strategic management. The study aims to elucidate the macro-level behaviors and features of firms, as well as their impact on firm performance.

The last underpinning theory of the study is the DC, which articulates the relationship between top management support as the mediator in the research and the relationship between cloud computing adoption and firm performance. The DC framework was developed by Teece et al. [31], as outlined in "Dynamic Capabilities and Strategic Management," which utilized the framework to examine the origins and techniques of wealth creation and acquisition by companies functioning in contexts of swift technological advancement [31].

The study will fill the gap of the TOE and DOI by including two firm-related theories and models, the theory of PBV and DC, to elucidate the connection between cloud computing adoption and firm performance, together with top management support and vendor support as mediating and moderating variables.

The research model

Theoretical framework

Figure 1 illustrates the theoretical framework for this study, which has been modified to align with the research objectives. The framework has four independent variables: technological context, inventive features, environmental context, and organizational context, with cloud computing adoption as the dependent variable. This research study employs a theoretical framework to examine whether top management support serves as a mediator in the relationship between independent variables and cloud computing adoption, while also assessing whether vendor support moderates the relationship between cloud computing adoption and firm performance.

Formulation of hypotheses

H1: Technological context is positively and significantly related to cloud computing adoption.

H1a: Cost-effectiveness is positively and significantly related to cloud computing adoption.

H1b: Security assurance is positively and significantly related to cloud computing adoption.

H2: Technological context is positively and significantly related to top management support.

H3: Innovative characteristics are positively and significantly related to cloud computing adoption.

H3a: Relative advantage is positively and significantly related to cloud computing adoption.

H3b: Complexity is negatively and significantly related to cloud computing adoption.

H3c: Compatibility is positively and significantly related to cloud computing adoption.

H4: Innovative characteristics are positively and significantly related to top management support.

H5: Environmental context is positively and significantly related to cloud computing adoption.

H5a: Industry influence is positively and significantly related to cloud computing adoption.

H5b: Competitive pressure is positively and significantly related to cloud computing adoption.

H5c: Regulatory compliance is positively and significantly related to cloud computing adoption.

H6: Environmental context is positively and significantly related to top management support.

H7: Organizational context is positively and significantly related to cloud computing adoption.

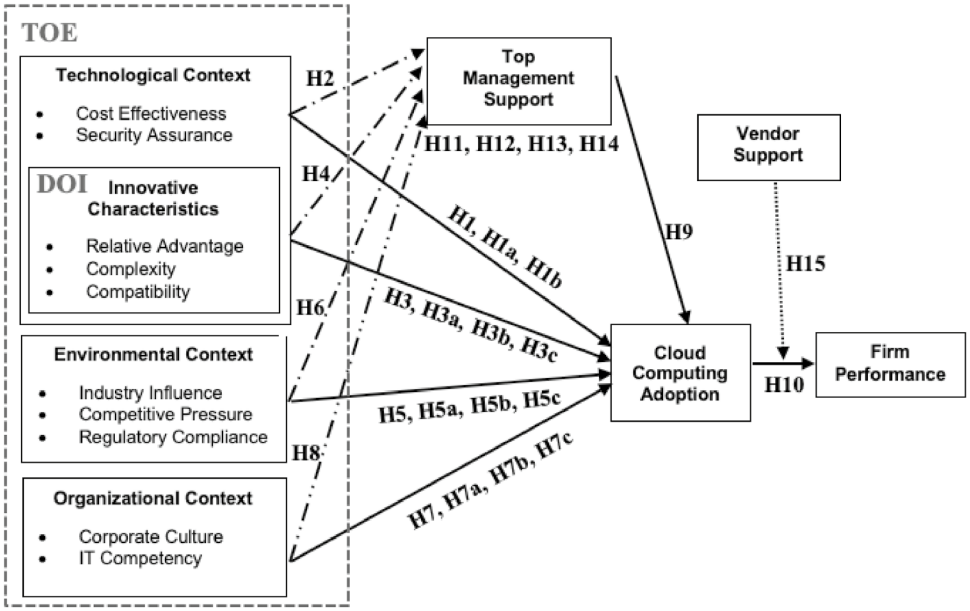
H7a: Corporate culture is positively and significantly related to cloud computing adoption.

H7b: IT competency is positively and significantly related to cloud computing adoption.

H8: Organizational context is positively and significantly related to top management support.

H9: Top management support is positively and significantly related to cloud computing adoption.

H10: Cloud computing adoption is positively and significantly related to firm performance.



Note: H: Indicates all the hypotheses of the study;
——▶ Indicates the hypothesized relationship (direct) between each construct
- - - ▶ Indicates the hypothesized relationship (direct) between each construct
·····▶ Indicates the hypothesized relationship (moderating)
H11–H14 Indicates the hypothesized relationship (mediating)

Fig. 1 Proposed theoretical framework

H11: Top management support mediates the relationship between technological context and cloud computing adoption.

H12: Top management support mediates innovative characteristics and adoption of cloud computing.

H13: Top management support mediates the relationship between environmental context and cloud computing adoption.

H14: Top management support mediates the relationship between organizational context and cloud computing adoption.

H15: Vendor support moderates the relationship between cloud computing adoption and firm performance such that the relationship becomes stronger as vendor support increases.

Technological context (H1) demonstrated a substantial impact on the adoption of cloud computing [32]. The participants often highlighted cost-effectiveness (H1b) in their discussions on the advantages of cloud computing [33]. Security (H1b) significantly influences the adoption of cloud computing, as identified in the study [34]. The findings of Li [16] show a positive connection between technology and management support (H2). All the sub-dimension hypotheses under innovative characteristics (H3) are supported throughout the empirical study [34]. Mohammed et al. [35] provided proof in support of the premise that relative advantage (H3a) will positively impact the decision to implement cloud ERP within SMEs. An empirical study on SMEs has validated that complexity (H3b) has a significantly positive influence on cloud computing adoption [34]. According to Omar et al.'s [36] study on cloud ERP systems, it has been proven that compatibility (H3c) significantly affects the adoption of cloud computing. Li's [16] study reveals a significant correlation between technology and management support (H4), with management support specifically denoting top management support, and technology refers to innovative characteristics. The hypothesis of environmental context (H5) demonstrated a considerable impact on the adoption of cloud computing [32]. Dutta and Mukherjee's [32] study has proven that industry (H5a) significantly influences the adoption of cloud computing, and the study encompassed all sectors. Competitive pressure (H5b) has been proven that it exerts a considerable influence on the adoption of cloud computing in a cross-industry analysis [32]. The findings of Amin et al. [37] substantiate the hypothesis of regulatory support (H5c) for cloud computing adoption. A study on logistics technology adoption, top management support was identified to mediate the relationship in the

environmental context (H6) [38]. The study by Kumar et al. [39] established a significant beneficial relationship between organizational context (H7) and the intention to utilize technology in organized retail outlets. Culture (H7a) is a significant influencing factor, as it reflects the community's stance on adopting cloud computing [40]. The study by Omar et al. [36] demonstrated that IT competencies (H7b) significantly influence the adoption of cloud computing. The organizational context (H8) and readiness exhibit a substantial positive relationship with top management support [41]. Teh et al. [42] conducted a systematic literature review indicating that top management (H9) significantly influences cloud computing adoption, corroborated by multiple prior research in this domain. Al-Sharafi et al. [7] conducted prior research on the factors influencing cloud computing integration and its effects on sustainable performance in SMEs, confirming that the adoption of cloud computing positively affects firm performance (H10). A study validated that top management support (H11, H12, H13, H14) mediates the innovative characteristics, technological, and organizational contexts of cloud computing adoption [41]. Vendor support (H15), as seen in Yu et al. [43] and Amoako-Gyampah et al. [44], plays a moderating role in supply chain viability, where vendor relationship with firm performance has been proven.

Research methodology

This section presents a comprehensive methodological framework designed to investigate cloud computing adoption determinants and performance outcomes among SMEs in Klang Valley, Malaysia. The selection of Klang Valley is analytically significant as it represents Malaysia's primary economic corridor, accounting for 46% of national GDP and hosting the highest concentration of technology-ready SMEs, making it an ideal context for examining early-stage cloud adoption patterns that may subsequently diffuse to other regions.

Research design and philosophy

The study employed a positivist management philosophy, the deductive approach method, and a descriptive and hypothesis testing design, utilizing a mono-method quantitative methodological choice utilizing a web-based survey and a cross-sectional study. While cross-sectional design limits causal inference, it provides a critical snapshot of current adoption factors during a pivotal digital transformation period for Malaysian SMEs.

Sampling strategy and justification

The study employed purposive sampling, specifically criterion sampling, targeting SMEs that met three criteria: (1) registered as SMEs per Malaysian government classification; (2) located within Klang Valley; and (3) currently using or evaluating cloud services. This non-probability approach was selected over random sampling for several reasons. First, the nascent stage of cloud adoption among Malaysian SMEs necessitates targeting organizations with relevant experience. Second, purposive sampling enables deeper insights into adoption factors among engaged organizations rather than superficial coverage of uninterested firms. Third, the approach aligns with exploratory research objectives in emerging technology contexts.

While purposive sampling limits statistical generalizability, analytical generalization remains valid through theoretical framework validation. The sample size of 350, though not statistically representative, exceeds minimum requirements for PLS-SEM analysis (10 times the maximum number of paths directed at any construct) and provides sufficient statistical power (0.80) for detecting medium effect sizes at $\alpha = 0.05$.

Data collection and validation

Pre-testing involved five respondents across different SME sectors to ensure question clarity, appropriate response scales, and completion time. Five pre-testing might not be sufficient in certain environment, but it was predetermined during the research design. Cognitive interviews were conducted to identify ambiguous items. The pilot study with 30 respondents enabled reliability assessment (all Cronbach's $\alpha > 0.7$) and preliminary factor structure validation.

Data cleaning procedures included outlier detection using SPSS (17 cases removed), missing value analysis (cases with > 10% missing removed), response pattern analysis to identify straight-lining and normality assessment using skewness and kurtosis thresholds (± 2).

Common method bias controls

While Harman's single-factor test yielded 48.6% variance (below but close to the 50% threshold), we acknowledge this test's limitations. Additional controls implemented include temporal separation between predictor and outcome variable collection, randomization of item order across constructs, use of different response formats for key variables, post-hoc marker variable analysis using social desirability scale, and unmeasured latent method factor in CFA showing < 25% variance.

Construct validity enhancement

Several constructs showed marginal reliability and validity metrics requiring remediation, corporate culture (CR = 0.758, AVE = 0.517) whereby items were refined through exploratory factor analysis, removing cross-loading items, cost-effectiveness (AVE = 0.582) whereby

conceptual overlap with relative advantage addressed through discriminant validity testing, and additional validation through nomological network assessment confirmed theoretical relationships.

Research results

This section concentrates on data analysis and its results, utilizing IBM SPSS and SmartPLS 4 for processing and analysis, with pertinent data displayed in tabular and graphical formats. Data screening, socioeconomic demographics, reliability, validity, normalcy, descriptive analysis, linearity, common method variance, multicollinearity, and correlation tests were examined using IBM SPSS software. Confirmatory tetrad analysis, internal consistency assessed through composite reliability, convergent validity evaluated via factor loadings and average variance extracted (AVE), discriminant validity determined using the Fornell–Larcker criterion, along with R-squared, F-squared, Q-squared, path coefficients for hypotheses, and tests for mediation and moderation were conducted using SmartPLS 4 software for both measurement and structural models.

Analysis

Before initiating the data collection, a reliability test was performed on 30 pilot cases to ascertain the degree of reliability, revealing that all variables from these cases exhibited a Cronbach's alpha exceeding 0.7. The subsequent phase involved data cleansing and analysis of the collected data. A total of 352 individuals participated in the questionnaire survey; however, 2 cases with missing data were eliminated, yielding 350 final cases for subsequent data analysis (Table 1).

The study employed Harman's single-factor test with IBM SPSS to evaluate common method variance, a critical aspect. As shown in Table 2, the outcome reveals that the variation is 48.6%, which is below 50%. Tehseen et al. [54] assert that common technique variance is not an issue, although value is at the bottom line, which could be further improved and will be discussed in the limitations in the last section. This discovery is crucial to guarantee the absence of bias, as data from both independent and dependent variables were collected from the same cohort of participants utilizing the identical instrumentation.

In contrast to AMOS, SmartPLS analysis necessitates the identification of a formative or reflective measurement model before proceeding with the analysis, as only the reflective measurement model is available in AMOS. To achieve this, confirmatory tetrad analysis in PLS (CTA-PLS) is utilized in SmartPLS [55], which requires at least four variables per construct; otherwise, the analysis will not be conducted. Based on Table 3, a zero-value straddle existed between the low and high adjustments of the confidence interval in each construct; therefore,

Table 1 Instruments

Section	Description	No. of items	Sources
A	Innovative Characteristics	15	Mohammed et al. [45], Ali et al. [46]
B	Technological Context	10	Qian et al. [47], Qasem et al. [48]
C	Environmental Context	14	Jaradat et al. [49], Qian et al. [47], Ali et al. [46]
D	Organizational Context	10	Alghushami et al. [50], Mohammed et al. [45]
E	Top Management Support	8	Qian et al. [47]
F	Vendor Support	5	Gangwar [51]
G	Cloud Computing Adoption	5	Wahsh and Dhillon [52]
H	Firm Performance	7	Garrison et al. [53], Gangwar [58]
I	Demographic	7	Author

Table 2 Common method variance

Component	Initial eigenvalues			Extraction sums of squared loadings		
	Total	% of var.	Cumulative %	Total	% of var.	Cumulative %
1	7.292	48.615	48.615	7.292	48.615	48.615
2	1.107	7.383	55.998	–	–	–
3	0.995	6.630	62.629	–	–	–
4	0.779	5.194	67.823	–	–	–
5	0.706	4.704	72.527	–	–	–

Table 3 Confirmatory tetrad analysis

	Constructs (1 set per construct)	*p* values	CI low adj	CI up adj
Relative advantage	RA2, RA3, RA4, RA5	0.287	−0.025	0.066
Compatibility	CA1, CA2, CA3, CA4	0.495	−0.035	0.066
Complexity	CX1, CX2, CX4, CX3	0.285	−0.041	0.112
Cost-effectiveness	CE1, CE2, CE4, CE3	0.223	−0.08	0.025
Security assurance	SC1, SC2, SC3, SC4	0.196	−0.031	0.107
Industry influence	II1, II2, II4, II3	0.747	−0.028	0.04
Competitive pressure	CP1, CP2, CP3, CP5	0.134	−0.023	0.11
Regulatory comp	RC1, RC2, RC3, RC4	0.409	−0.058	0.123
Corporate culture	CC1, CC2, CC3, CC4	0.197	−0.143	0.04
IT competency	IC1, IC2, IC3, IC4	0.345	−0.044	0.105
Top Mgmt. support	TMS1, TMS2, TMS3, TMS4	0.231	−0.034	0.085
Vendor support	VS1, VS2, VS3, VS4	0.136	−0.012	0.057
Cloud Com. adoption	CCA1, CCA2, CCA4, CCA3	0.737	−0.029	0.040
Firm performance	FP1, FP2, FP3, FP5	0.242	−0.015	0.042

all these represent reflecting model evaluations [55, 56]; only one set per construct is shown in Table 3 as most of the constructs have more than five variables; hence, there are few combinations' results available in CTA. This is also aligned with previous studies with reflective measurement models, such as Alismaili et al. [57] and Qasem et al. [48]. With the reflective measure model, all the indicators will point out from the variables toward indicators instead of the other way around, and such a direction of the indicators was established as the default in the rest of the tests in SmartPLS.

The composite reliability coefficient should exceed 0.7 to signify sufficient convergence or internal consistency [58, 59]. The composite reliability results in Table 4 indicate that all figures above 0.7, demonstrating sufficient dependability of the instruments, which is a criterion for ensuring robust and reliable study findings.

With the reflective model assessment identified in the study, average variance extracted (AVE) will be applied to the extent to which the construct or latent variable can account for the variation in the items (Analysis [60]), whereby an AVE of at least 0.50 is highly recommended for adequate convergence [61]. Based on the results obtained after removing four questions (2 from corporate culture), as shown in Table 5, all variables' AVEs exceed 0.50, surpassing the acceptable threshold.

The findings of the hypotheses evaluated using path coefficient to obtain the *p* value are presented in Table 6. There are 20 direct hypotheses based on the study's theoretical framework, excluding those involving mediating and moderating effects; 15 hypotheses are supported out of 20, based on the study's results, while 5 are not supported. All key elements of the study's theoretical framework—the technological context (H1), innovative characteristics (H3), environmental context (H5), and organizational context (H7)—exhibit a positive and substantial relationship with cloud computing adoption. The hypothesis (H10) indicating that cloud computing positively impacts firm performance is supported.

Table 4 Composite reliability

Variable	Composite reliability
Relative advantage	0.882
Complexity	0.911
Compatibility	0.909
Cost-effectiveness	0.873
Security assurance	0.929
Industry influence	0.952
Competitive pressure	0.903
Regulatory compliance	0.876
Corporate culture	0.758
IT competency	0.920
Top management support	0.940
Vendor support	0.956
Cloud computing adoption	0.919
Firm performance	0.946

Table 7 delineates the outcomes of the mediation analysis conducted using SmartPLS 4, encompassing a total of four hypotheses structured according to the theoretical framework outlined earlier. The study will not adhere to the Baron and Kenny [62] methodology for mediation analysis, which involves examining direct, indirect, and total effects and then categorizing the mediation

Table 5 Average variance extracted (AVE)

Variable	Average variance extracted (AVE)
Relative advantage	0.601
Complexity	0.672
Compatibility	0.666
Cost-effectiveness	0.582
Security assurance	0.724
Industry influence	0.831
Competitive pressure	0.653
Regulatory compliance	0.588
Corporate culture	0.517
IT competency	0.699
Top management support	0.663
Vendor support	0.814
Cloud computing adoption	0.693
Firm performance	0.746

as full, partial, or no mediation. Instead, it will employ the Preacher and Hayes [15] approach, which utilizes bootstrapping to estimate the indirect effect. Based on Preacher and Hayes [15], a mediation effect is present if the 95% Bootstrap Confidence Interval (CI) does not straddle 0 between the lower and upper limits. Among all four related mediation hypotheses, one hypothesis has no mediation effect: top management support does not mediate the relationship between technological context and cloud computing adoption (H11).

Table 8 is the result of the moderating analysis conducted using SmartPLS 4 with PLS analysis. A simple slope analysis in SmartPLS 4 can obtain the moderation analysis result. The p value is below 0.05, signifying statistical significance; thus, the hypothesis is corroborated. Based on Fig. 2 with simple slope analysis, the conditional $X–Y$ (cloud computing adoption–firm performance) slope becomes steeper as the values of Z (vendor support)

Table 6 Hypothesis testing

Hypothesis	β/Ori. sample	Sample mean	Std. deviation	*T* statistics	*P* values	Outcome
H1	0.684	0.678	0.165	4.141	0.000	Supported
H1a	0.054	0.056	0.055	0.979	0.164	Not supported
H1b	0.178	0.176	0.053	3.351	0.000	Supported
H2	−0.304	−0.311	0.212	1.430	0.076	Not supported
H3	−0.346	−0.340	0.165	2.097	0.018	Supported
H3a	0.211	0.211	0.058	3.651	0.000	Supported
H3b	0.042	0.040	0.071	0.595	0.276	Not supported
H3c	−0.030	−0.027	0.061	0.499	0.309	Not supported
H4	0.419	0.426	0.200	2.092	0.018	Supported
H5	0.150	0.154	0.059	2.554	0.005	Supported
H5a	0.105	0.101	0.059	1.779	0.038	Supported
H5b	0.097	0.098	0.059	1.641	0.050	Supported
H5c	0.034	0.038	0.046	0.730	0.233	Not supported
H6	0.191	0.191	0.066	2.905	0.002	Supported
H7	0.103	0.103	0.059	1.753	0.040	Supported
H7a	0.132	0.133	0.048	2.738	0.003	Supported
H7b	0.189	0.186	0.045	4.190	0.000	Supported
H8	0.547	0.549	0.057	9.682	0.000	Supported
H9	0.367	0.363	0.055	6.672	0.000	Supported
H10	0.783	0.785	0.026	29.906	0.000	Supported

Table 7 Mediation analysis

Hypotheses	Path coefficient	*T* value	*p* value	BI [2.5%; 97.5%]	Mediation
H11	−0.111	1.391	0.164	−0.291; 0.028	No
H12	0.154	1.986	0.047	0.020; 0.328	Yes
H13	0.070	2.664	0.008	0.025; 0.126	Yes
H14	0.201	5.261	0.000	0.133; 0.284	Yes

Table 8 Moderating analysis

Hypothesis	*β*	Sample mean	Std deviation	*T* statistics	*p* values
H15	0.073	0.074	0.021	3.501	0.000

increase, and vendor support moderates the positive relationship (accentuating) between cloud computing adoption and firm performance, such that the relationship becomes stronger as vendor support increases. Vendor support moderates the positive relationship between cloud computing adoption and firm performance, such that the relationship becomes stronger as vendor support increases.

Discussion

The study's findings reveal a nuanced picture of cloud adoption among Malaysian SMEs that both confirms and challenges existing theoretical assumptions.

Theoretical implications

This study advances technology adoption literature through three key contributions: first, the integrated TOE-DOI-PBV-DC framework demonstrates theoretical complementarity rather than redundancy. While TOE and DOI effectively explain adoption antecedents ($R^2 = 0.684$ for technological context), PBV and DC theories illuminate performance outcomes ($R^2 = 0.635$). This multi-theoretical approach addresses criticisms of "theory shopping" by showing how each theory contributes unique explanatory power at different stages of the adoption-performance chain. TOE framework clearly explained the process of intra-firm innovation adoption, while the DOI model offers a more thorough elucidation of the innovative dynamics associated with technology adoption, integrating both ideas can yield a more thorough and supplementary comprehension of the IT adoption phenomenon.

Second, the non-significance of traditionally important factors (cost-effectiveness, complexity, compatibility) in the Malaysian context challenges the universal applicability of Western-developed adoption models. This finding suggests that contextual factors—including development stage, government support structures, and cultural dimensions—moderate the influence of adoption determinants, necessitating localized theoretical refinements. The findings could be due to different kinds of research methodology with different methods and instruments, as well as different target populations such as geographic, segment, industry, sample size, and others. These findings address the evidence gap as identified in the study's problem statement and also address the research questions of the study.

Third, the mediating role of top management support (explaining 41.9% of indirect effects) extends upper echelons theory by demonstrating how leadership translates external pressures and internal capabilities into adoption

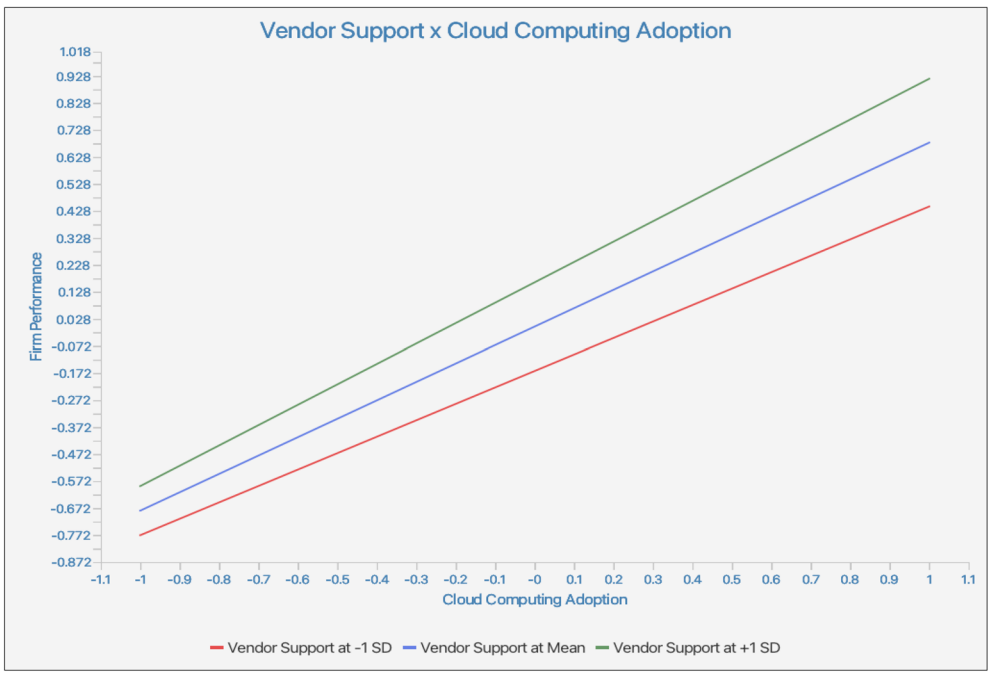


Fig. 2 Moderating simple slope analysis

decisions. This finding enriches our understanding of micro-foundations in organizational technology adoption. Both of these top management support and vendor support findings address the theoretical gap as identified in the study's problem statement and also address the research questions of the study that top management support mediates the relationship between TOE contexts and cloud computing adoption, while vendor support moderates the relationship between cloud computing adoption and firm performance.

Contextual insights

The Malaysian SME context reveals several distinctive patterns: cost-effectiveness insignificance ($\beta = 0.054$, $p = 0.164$) contradicts findings from developed markets but aligns with emerging economy studies where capability-building trumps cost considerations. Malaysian government subsidies and vendor promotional strategies may temporarily mask cost concerns, suggesting a potential future shift as markets mature.

Complexity non-significance ($\beta = 0.042$, $p = 0.276$) indicates a concerning capability gap—SMEs may lack the technical sophistication to evaluate system complexity, making them vulnerable to inappropriate technology choices. This finding underscores the need for enhanced digital literacy initiatives.

The strong influence of competitive pressure ($\beta = 0.097$, $p = 0.050$) and industry influence ($\beta = 0.105$, $p = 0.038$) reflect the collectivist nature of Malaysian business culture, where peer adoption strongly influences technology decisions.

Practical implications

Based on empirical findings, we provide specific, actionable recommendations: for SME managers to develop structured IT competency programs focusing on cloud architecture understanding rather than just usage skills, establish cloud adoption committees combining IT and business unit representatives to ensure alignment and negotiate collective bargaining agreements with vendors through industry associations to improve terms.

For cloud vendors, to design tiered support packages matching SME maturity levels (basic, intermediate, advanced), provide sandbox environments enabling risk-free experimentation addressing complexity concerns and develop industry-specific templates reducing customization requirements.

For policymakers, to implement sector-specific incentives (e.g., manufacturing vs. services) rather than blanket policies, establish cloud competency certification programs addressing the identified capability gaps and create vendor evaluation frameworks helping SMEs assess security and reliability claims.

Conclusion

This study provides empirical evidence of cloud computing adoption determinants and performance outcomes among Malaysian SMEs, contributing to both theoretical advancement and practical application. The integrated multi-theory framework demonstrates how different theoretical lenses complement rather than compete in explaining complex technology adoption phenomena. The identification of context-specific factors challenges universal adoption models and highlights the need for localized theoretical development.

The strong adoption–performance relationship ($R^2 = 0.635$) validates cloud computing's strategic value for SMEs, while the mediating and moderating roles of top management and vendor support provide actionable insights for accelerating adoption. As Malaysia pursues its digital transformation agenda, these findings offer evidence-based guidance for stakeholders seeking to enhance SME competitiveness through cloud technologies.

This study has several limitations requiring acknowledgment. In terms of methodological limitations, purposive sampling limits generalizability to the broader SME population, cross-sectional design prevents causal inference and temporal dynamics assessment, self-reported measures introduce potential response bias despite control mechanisms and single-region focus (Klang Valley) may not represent nationwide patterns. In terms of measurement limitations, marginal reliability for some constructs (corporate culture CR = 0.758) suggests measurement refinement needs, common method variance (48.6%) while acceptable, remains concerning and pilot testing with 30 respondents may be insufficient for robust validation. The study employed online survey, which makes it impossible or unlikely to hear from participants about factors that matter to them or to obtain further clarification or feedback from them.

Future research should address these limitations through longitudinal designs tracking adoption evolution and performance impacts over time, multi-level analysis incorporating individual, organizational, and environmental factors, comparative studies across ASEAN countries identifying regional versus country-specific factors, mixed-methods approaches combining surveys with in-depth case studies for richer insights and industry-specific investigations recognizing sectoral differences in adoption patterns. Probability sampling techniques could enhance the robustness of the data analysis and allowing for generalization to represent Malaysian SMEs. The instruments of future studies could be revised to enhance the quality and provide more solid measurement findings, which have been identified as limitations in the study with the bottom-line figures.

Author contributions
We conceptualized the study, developed the research framework, and supervised the project. We designed the survey instrument, coordinated data collection, and performed the statistical analyses. We contributed to the literature review, data interpretation, and drafting of the manuscript. We critically reviewed and revised the manuscript, approved the final version for submission, and agreed to be accountable for all aspects of the work.

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Data availability
The dataset and survey instrument are available (please email the author) for replication purposes. Access restrictions apply to protect participant confidentiality.

Declarations

Ethics approval and consent to participate
This research received ethical approval from the Universiti Putra Malaysia Ethics Committee (Reference: UPM/TNCPI/RMC/1.4.18.2/F2, Approval Date: 14 December 2022). All participants provided informed consent before participation.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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